

EXCALIBUR

DESIGN & ENGINEERING PORTFOLIO

STEM Racing Panama · Development Class · 2026

The Oxford School · Car: Arthur 1

Manuel Gomes — Manufacturing Eng.

Giancarlos Pan — Design Eng.

Nabil Manjara — Design Eng.

Roberto Ho — Design Eng.

Heidy Qiu — Graphic Designer

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01 Research & Design Concepts

Why we made the decisions we made — the science, the rules, and our first ideas.

Design brief & objectives

TARGET MASS	61.5 g without the CO2 cartridge — a 1.5 g margin above the 60.0 g minimum.
PRIORITIES	1. Regulatory compliance · 2. Air resistance · 3. Manufacturability
OUR BRIEF	A reliable car that is practical to manufacture and runs with low air resistance, with the official halo, standard running gear and front and rear wings. Success is judged by checking it against the technical regulations.

Aerodynamics & physics research

DRAG & SHAPE	A moving car pushes air out of its way, creating drag. A body that narrows towards the back helps air flow around it; a teardrop has “minimum form drag” (NASA report 19790021981, Sec. VII).
BOUNDARY LAYER	A thin layer of air forms next to the surface. Smooth surfaces reduce skin-friction drag; pressure differences cause form drag and flow separation (NASA Glenn, Beginner’s Guide to Aeronautics).

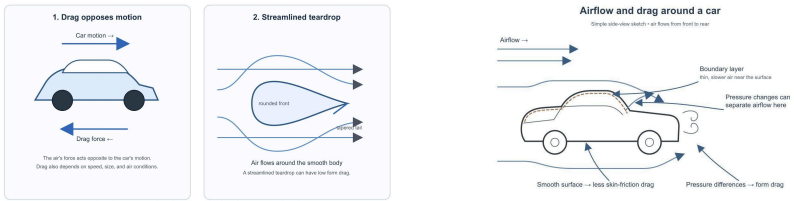


Fig. 1.1 Drag and streamlining · airflow and boundary layer around a car

Concept selection

WHY V3 WENT FORWARD	More accessible CNC body, three thin front wings, and almost the same drag force.
CARRIED TO V5	Same wing shape, but thicker and further from the wheels — giving less drag.

Regulation analysis

Rule	Requirement
D3.4	Total length: min 170 mm / max 210 mm
D3.5	Total width: max 90 mm
D3.6	Total weight: min 60.0 g without cartridge (weigh after finishing)
D4.2	Preserve the official no-go-zone
D4.3.3	Halo notch 34.0 ± 1.0 mm above track; axle centres 11.0 mm from body underside
D5.1	Inserted cartridge rear centre 30.0–40.0 mm above track
D6.1–D6.5	Supplied running gear; wheel visibility; 3 mm exclusion zone; max 20 mm front obstruction; free rotation and four-wheel contact
D7.6 / D8.6	Wings: span min 60 mm, chord 15–30 mm, thickness 5–15 mm
D8.5 – D9.2	Rear wing underside ≥ 35 mm; two closed tether guides within 15 mm of axle centres, bore 3.5–6 mm; 200 g test

Concept evaluation

Concept	Drag	Mass	Manuf.	Regs	Looks	Total
1	1	5	5	1	2	2.8
2	3	3	2	2	3	2.6
3	4	4	3.5	5	5	4.3
4	5	5	4	4.3	5	4.66
5	5	5	3.8	5	5	4.76

Initial sketches

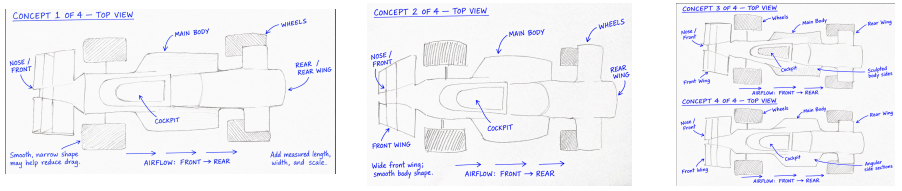


Fig. 1.2 Freehand concept sketches, annotated (top view)

02 Design Development & CAD

From the no-go-zone to the final model — every iteration, with a reason for every change.

CAD workflow

SOFTWARE	Autodesk Fusion 360, Ansys Discovery, Excel
AXLE HOLES	11.0 mm from the underside of the body (D4.3.3)
RUNNING GEAR	Standard wheels, axles and bushes inserted as supplied parts — geometry unmodified (D6.1)

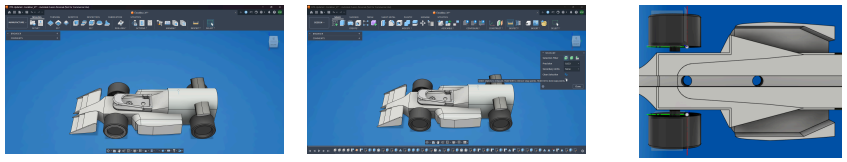


Fig. 2.1 CAM evidence · CAD evidence (D3.1.1) · axle-hole position

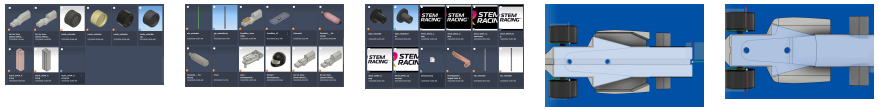


Fig. 2.2 Supplied running gear · assembly with the no-go-zone in section

Front wing development

SPAN / CHORD / THICKNESS	34 / 17 / 6 mm (D7.6) — entirely ahead of the front wheel centre line (D7.4), unobstructed in front view (D7.5)

Rear wing & body development

SPAN / CHORD / THICKNESS	63 / 17.2 / 3.6 mm (D8.6) — underside 43 mm above track (D8.5), behind the rear wheel centre line (D8.4)
BODY FEATURES	Canister loft, sidepods, underbody, nose — what each does:

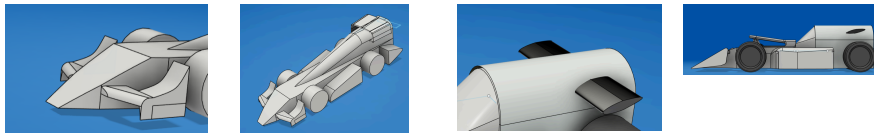
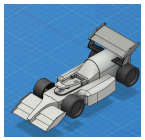


Fig. 2.3 Front wing iterations · rear wing / sidepod development

Mass strategy

PREDICTED CAD MASS	57.7 g (foam + wheels 3.5–3.8 g each + axles 4 g each) + 2.5 g for paint, filler and decals

Design iterations



Arthur 1 – V1 — 03/07/2026

WHAT CHANGED Everything was modified.
WHY To comply with the regulations.



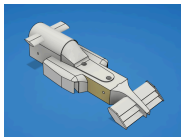
Arthur 1 – V2 — 16/07/2026

WHAT CHANGED Shape of the car — more streamlined.
WHY To comply with the regulations.
RESULT Original design, but more drag force.



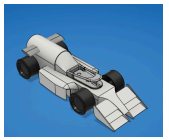
Arthur 1 – V3 — 17/08/2026

WHAT CHANGED Shorter wing length and span.
WHY To reduce the drag force.
RESULT Minor differences in drag force.



Arthur 1 – V4 — 16/09/2026

WHAT CHANGED Front axle moved forward 6 mm; wings unchanged.
WHY The halo was too close to the axle.
RESULT Wheels now too close to the front wings.



Arthur 1 – V5 (final) — 28/09/2026

WHAT CHANGED More space between wheel and wing; thicker wings.
WHY The 3D-printed wings were very fragile.
RESULT Less drag force on the car.

Development timeline

Arthur 1 – V1	Arthur 1 – V2	Arthur 1 – V3	Arthur 1 – V4	Arthur 1 – V5
CREATED July 3	CREATED July 16	CREATED August 17	CREATED September 16	CREATED September 29
SIMULATED July 3	SIMULATED July 16	SIMULATED August 17		SIMULATED September 29

03 Analysis & Testing

How we tested every iteration under the same virtual conditions.

CFD / virtual wind tunnel setup

SOFTWARE	Ansys Discovery
INLET VELOCITY	20 m/s — we used 72 km/h as our average race speed, which converts to 20 m/s.
ITERATIONS	300
BOUNDARY CONDITIONS	Non-translating ground, inlet and outlet flow, plastic solid
WHY IDENTICAL CONDITIONS?	So we could compare the drag force of each car fairly.

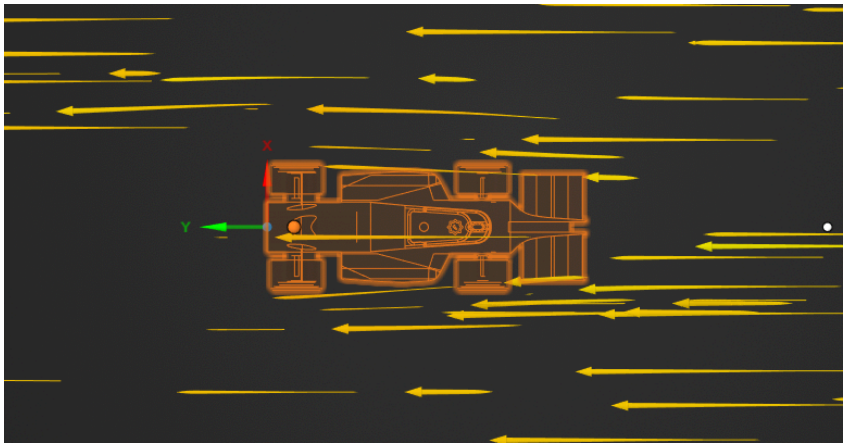


Fig. 3.1 CFD domain / mesh

Comparison of iterations

Iteration	Cd	Drag force (N)	Mass (g)
Iteration 1	0.0284	0.310	53.9
Iteration 3	0.0170	0.384	60.0
Iteration 5	0.0170	0.372	60.3

CFD results

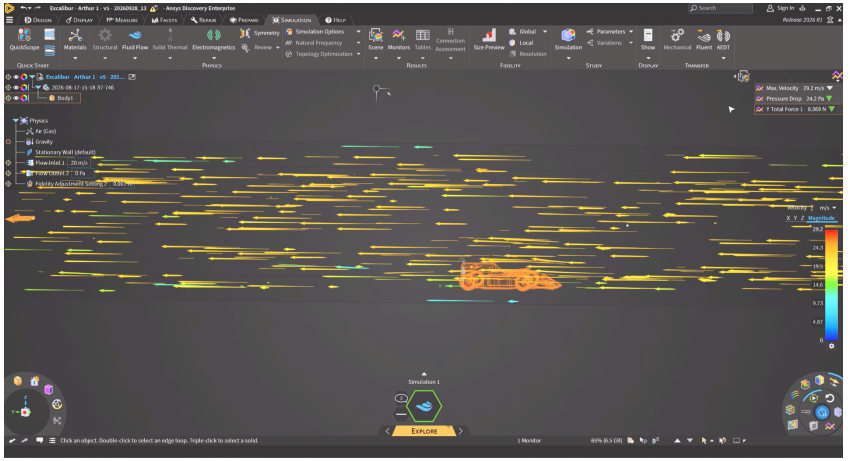


Fig. 3.2 CFD result — flow and pressure

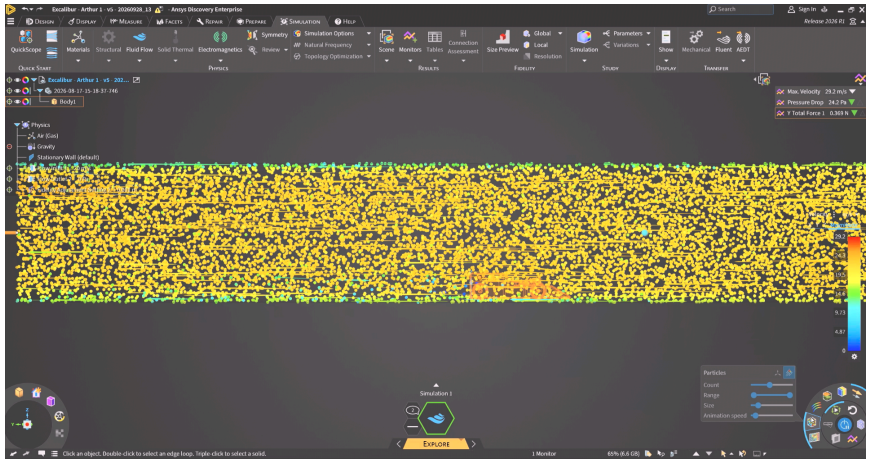


Fig. 3.3 CFD result — comparison view

04 Manufacture — CAM, CNC & Assembly

How the digital model became a legal, race-ready car — with quality control at every stage.

Wings & components

METHOD	3D printed
MATERIAL AND WHY	PLA — easy to print with minimal warping on a Bambu Lab A1.
PRINT SETTINGS	0.4 mm nozzle, standard 0.20 mm profile

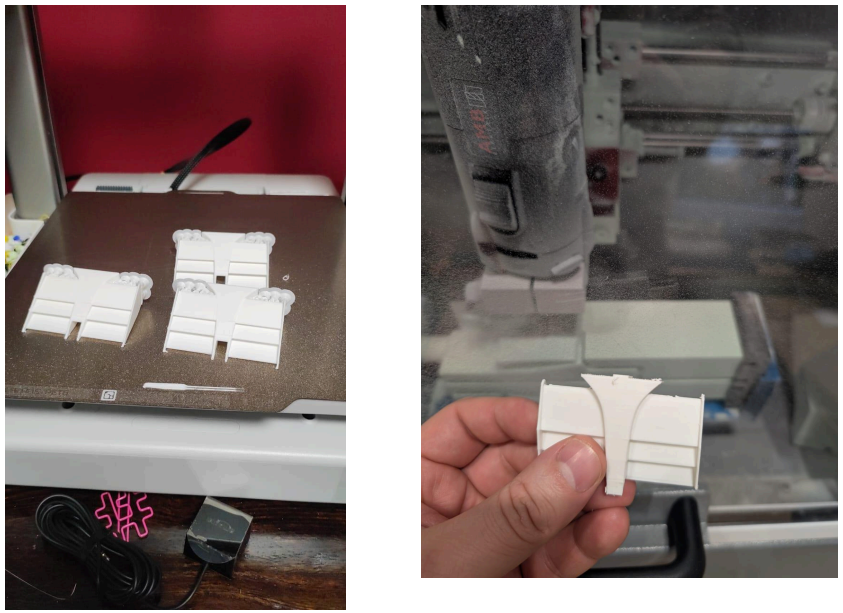


Fig. 4.1 Wing manufacture photos and slicer settings

Finishing & assembly

PAINT MASS LOG	Before 61.8 g → after 60.2 g
RUNNING GEAR	Wheels, bushes and axles fitted; all four wheels spin freely (D6.5) and touch the track together (D6.4)
DECALS AND ID	STEM Racing decal on both sides between the wheels (D4.5); Car A / Car B identified (D1.14)
CARTRIDGE CHAMBER	Clear of axles and screw eyes (D5.2)

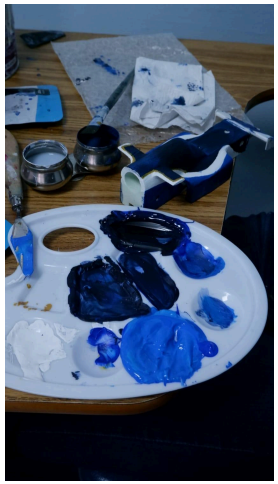


Fig. 4.2 Assembly and paint

Quality control

FINAL MASS	Car A: 60.2 g · Car B: 60.3 g
FINAL SIZE	Length 183 mm · width 63 mm

05 Drawings, Compliance & Evaluation

Proof of compliance, the finished car, and an honest evaluation.

Engineering drawings (D1.15)

VIEWS	Front, side and plan views plus an isometric, with all relevant dimensions and tolerances labelled with the regulation
MATERIALS & TITLE BLOCK	Material for every component; scale, projection symbol, drawn-by and date

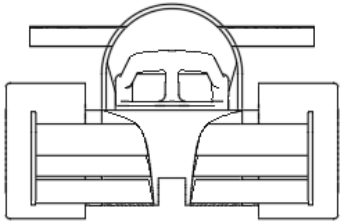


Fig. 5.1 Orthographic drawing (Fusion 360)

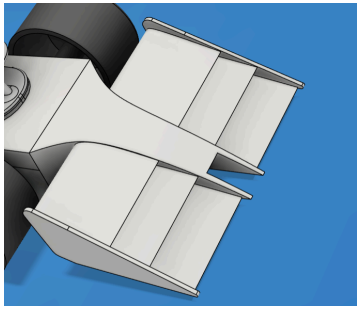


Fig. 5.2 Wing sections, dimensioned

Wing identification (D7.2 / D8.2)

FRONT WING	Span 34 mm · chord 17 mm · thickness 6 mm
REAR WING	Span 63 mm · chord 17.2 mm · thickness 3.6 mm



Fig. 5.3 The finished car — Arthur 1

Compliance summary

Rule	Measurement	Ours	Allowed
D3.4	Length	183 mm	170–210
D3.5	Width	63 mm	max 90
D3.6	Mass	60.1 g	min 60.0
D4.3.3	Halo notch	34 mm	34.0 ± 1.0
D5.1	Cartridge height	38 mm	30.0–40.0
D6.3	Front wheel obstruction	9.7 mm	max 20
D7.6	Front wing span / chord / thickness	34 / 17 / 6 mm	
D8.5	Rear wing height	39 mm	min 35
D8.6	Rear wing span / chord / thickness	63 / 17.2 / 3.6 mm	
D9.2	Tether guide bore	6 mm	3.5–6

Evaluation

WHAT WORKED BEST	Through the process of continuously finding and fixing our mistakes, we managed to improve the car and meet every regulation.
WHAT WE WOULD CHANGE	After each trial, we could have done more detailed notes on every version's changes and test results.
WITH MORE TIME WE WOULD...	Build a sixth version of the car applying what we learned from our earlier mistakes, then test it against our final car (version 5) to compare designs and have an option to choose from.